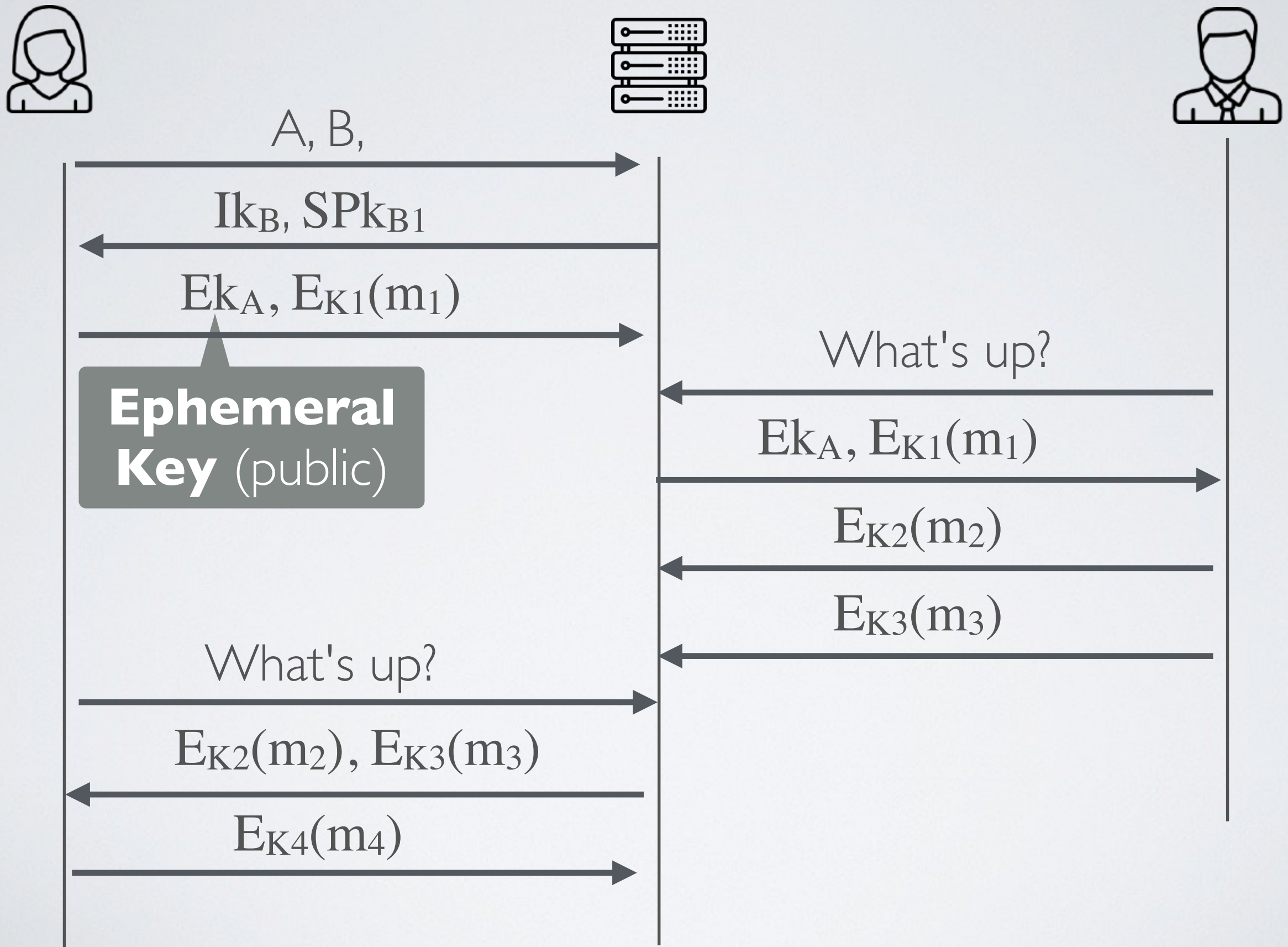


# Opening a channel between Alice and Bob



# Triple Diffie-Hellman Key Exchange (3DH)



$Ik_A$

$ECDH_1$

$ECDH_2$

$Ik_B$



$Ek_A$

$ECDH_3$

$SPk_B$

$ECDH(Ik_A, SPk_B)$

$ECDH(Ek_A, SPk_B)$

$k_{root} = HKDF( ECDH_1, ECDH_2, ECDH_2 )$

$ECDH(Ek_A, Ik_B)$